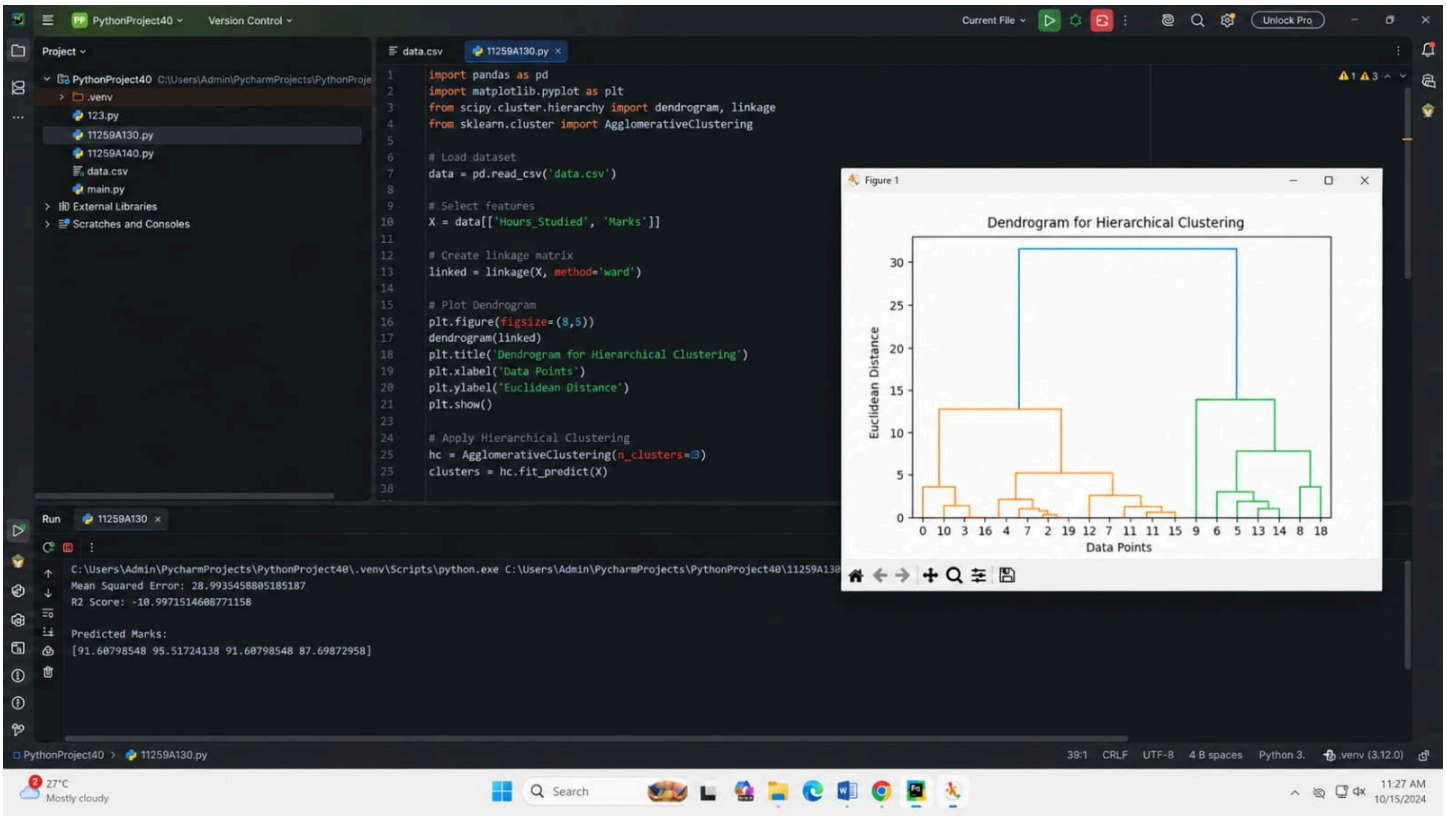
Experiment 6: Implement Hierarchical Clustering

Aim

To perform hierarchical clustering on a dataset using the Agglomerative Clustering algorithm, visualize the hierarchical relationships using a dendrogram, and group students into three clusters based on their study hours and marks.

Algorithm

1. Import the required libraries: Pandas, Matplotlib, dendrogram, linkage, and AgglomerativeClustering.
2. Load the dataset from the CSV file data.csv using pd.read_csv().
3. Select the Hours_Studied and Marks columns as input features.
4. Create a linkage matrix using linkage(X, method='ward') to calculate the hierarchical relationships between data points.
5. Plot the dendrogram using dendrogram(linked) to visualize the merging of clusters.
6. Set the graph title, axis labels, and figure size, and display the dendrogram.
7. Create an Agglomerative Clustering model with three clusters using AgglomerativeClustering(n_clusters=3).
8. Apply the clustering algorithm using hc.fit_predict(X) to assign cluster labels to each data point.
9. Add the cluster labels as a new column named Cluster to the original dataset.
10. Display the updated dataset.
11. Create a scatter plot of Hours_Studied against Marks, using different colors to represent the assigned clusters.
12. Add axis labels and a title, and display the scatter plot.



Result

The hierarchical clustering algorithm successfully grouped the dataset into three clusters based on students' study hours and marks.